

CET104 – UFCD5106

Projeto Final Redes de Computadores – Implementação de Rede Corporativa NETCORP



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Índice

Conteúdo

Projeto Final Redes de Computadores – Implementação de Rede Corporativa NETCORP.....	1
Introdução.....	3
Objetivos.....	3
Enunciado / Exercícios	4
Topologia	6
Endereçamento de IPs.....	7
1) VLANs, VTP e Etherchannel.....	9
2) Redundância com HSRP	12
3) Roteamento dinâmico e Redistribuição.....	13
4) NAT (PAT) em todos os routers.....	15
5) ACLs – Listas de Controlo de Acesso.....	17
6) Segurança de Camada 2	18
7) Autenticação AAA, IEEE 802.1X e SSH	19
8) VPN – Virtual Private Network	25
9) Teste e Validação	26
Bibliografia	30
Conclusão	30

Introdução

Este projeto implementa uma rede corporativa segura e redundante para a empresa NetCorp no Cisco Packet Tracer.

A rede interliga um Data Center a três filiais, garantindo comunicação interna, acesso à Internet e gestão centralizada de serviços. A topologia aplica conceitos de routing, switching e segurança aprendidos no curso.

A solução combina VLANs, HSRP, múltiplos protocolos de roteamento (OSPF, EIGRP, RIP) com redistribuição, NAT, ACLs e autenticação AAA/802.1X.

Toda a configuração foi validada através de testes de conectividade e failover.

Objetivos

- Implementar switching seguro com VLANs, VTP, EtherChannel e segurança de camada 2 (Port Security, DHCP Snooping).
- Garantir redundância com HSRP no Data Center e roteamento dinâmico (OSPF, EIGRP, RIP) com redistribuição.
- Assegurar segurança com ACLs, NAT, autenticação AAA/802.1X e SSH.
- Fornecer serviços centrais (DHCP, DNS, VPN) via servidor SRV1.
- Validar e documentar a implementação com testes extensivos.

Enunciado / Exercícios

A empresa NetCorp expandiu suas operações e agora possui um Data Center e três filiais interligadas.

Os técnicos de rede devem implementar a infraestrutura de comunicação e segurança que suporte os serviços internos, acesso à Internet e políticas de gestão.

A topologia inclui:

- Dois routers no data center (R1 e R2) – Responsáveis pela redundância e redistribuição de rotas;
- Três filiais (A, B e C) com routers próprios (R3, R4 e R5) e switches de acesso;
- Serviço central (SRV1) com DHCP, DNS, TACACS+ e Syslog;
- Rede de Internet (ISP) para ligação externa.

Rede	Endereço	Máscara	Função
Backbone R1–R2	10.0.0.0/30	/30	HSRP
Filial A	10.0.1.0/24	/24	OSPF
Filial B	10.0.2.0/24	/24	EIGRP
Filial C	10.0.3.0/24	/24	RIP
VLAN 10	192.168.10.0/26	/26	Administração
VLAN 20	192.168.20.0/26	/26	Vendas
VLAN 30	192.168.30.0/26	/26	Financeiro
VLAN 99	192.168.99.0/30	/30	Gestão
Rede ISP	200.10.10.0/30	/30	Internet

Etapas de configuração:

1) VLANs, VTP e Etherchannel

- Criar as VLANs 10 (Admin), 20 (Vendas), 30 (Financeiro) e 99 (Gerenciamento).
- Configurar o SW1 como VTP Server e os restantes switches como VTP Clients.
- Criar EtherChannel (LACP) entre SW1 e SW2.

2) Redundância com HSRP

- Implementar HSRP entre os routers R1 e R2.
- R1 → prioridade 110 (ativo).
- R2 → prioridade 100 (standby).
- Testar failover desligando o router principal.

3) Roteamento dinâmico e Redistribuição

- R3 (Filial A) → OSPF área 0.
- R4 (Filial B) → EIGRP AS 100.
- R5 (Filial C) → RIP versão 2.
- R1 ou R2 deve redistribuir as rotas entre OSPF, EIGRP e RIP.

4) NAT (PAT) em todos os routers

- Implementar o NAT overload (PAT) em todos os routers (R1 a R5).
- As interfaces LAN devem ser inside, e as interfaces para o ISP, outside.
- Confirmar acesso à internet (ping à interface loopback ou IP do ISP).

5) ACLs – Listas de Controlo de Acesso

- Acesso ao servidor (R1) – Apenas a VLAN 10 pode aceder ao servidor TACACS+ e DNS.
- Restrição Inter-VLAN – Impedir a comunicação direta entre a VLAN 20 e VLAN 30.
- Gestão (SSH) – Apenas a VLAN 10 pode aceder via SSH aos equipamentos.

6) Segurança de Camada 2

- Port-Security – Aplicar o Port-Security em todas as portas de acesso a PCs, deve permitir apenas um mac-address, o mesmo deve ser aprendido dinamicamente e caso ocorra alguma violação deverá desligar imediatamente a porta.
- DHCP Snooping – Ativar o DHCP Snooping, marcar as portas como trusted e untrusted.
- Dynamic ARP Inspection – Ativar o DAI, utilizando a base dados do DHCP Snooping.

7) Autenticação AAA, IEEE 802.1X e SSH

- Configurar autenticação centralizada via TACACS+/RADIUS no servidor SRV1.
- Ative o protocolo de controlo de acesso à rede baseado em porta, configurando um servidor AAA para decidir se um dispositivo pode aceder à rede. (Users à escolha, password cinel).
- Ativar SSH em todos os equipamentos e desativar o telnet.

8) VPN – Virtual Private Network

- Uma das máquinas de um LAN deve aceder através de uma VPN de acesso remoto a outra rede.

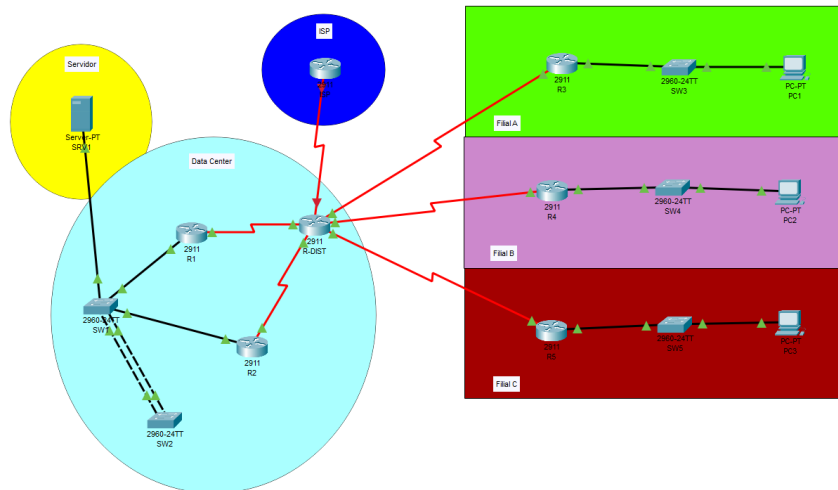
9) Teste e validação

- Testar ping entre filiais.
- Verificar redistribuição de rotas com o comando show ip route.
- Validar NAT com o comando show ip nat translations.
- Confirmar ACLs e segurança de portas (show access-lists, show port-security).
- Testar AAA e SSH.

Para guardar as configurações vou efetuando o comando “wr” periodicamente.

Topologia

Sendo que este trabalho oferece uma maior liberdade ao criar a topologia, decidi ir com este formato em estrela.

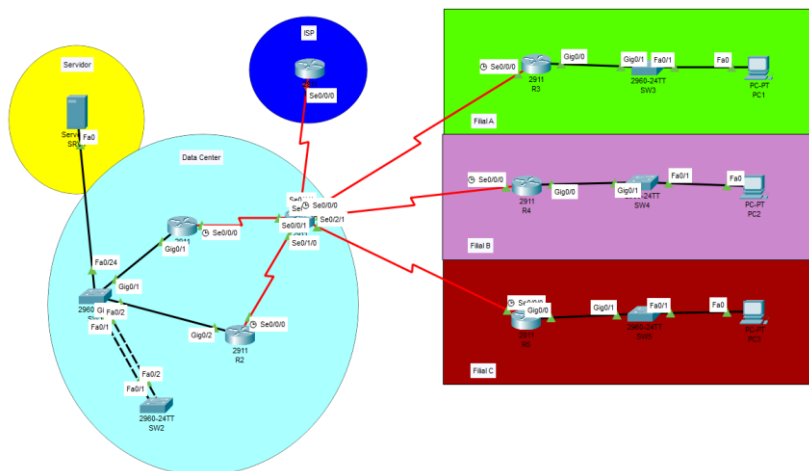


Decidi implementar um router central “R-DIST” para centralizar a redistribuição e simplificar a gestão de rotas. Em cada exercício vou adicionar as configurações necessárias para este router funcionar corretamente.

Pontos-chave da topologia:

- R-DIST é o hub central (estrela)
- Todos routers (R1, R2, R3, R4, R5) conectam-se ao R-DIST
- Data Center tem R1 e R2 ambos conectados ao SW1
- SW2 só serve para EtherChannel com SW1 (não conecta a routers)
- Cada filial tem o seu router conectado diretamente ao R-DIST

A topologia vai ter estas ligações:



Adicionalmente vou editar o hostname de cada dispositivo para ser igual aos nomes na topologia.

Endereçamento de IPs

Vou agora atribuir IPs de acordo com a tabela presente no enunciado.

R-DIST:

```
Router(config)#hostname R-DIST
R-DIST(config)#
R-DIST(config)#
R-DIST(config)#
R-DIST(config)#interface serial0/0/0
R-DIST(config-if)#description CONEXAO-INTERNET-ISP
R-DIST(config-if)#ip address 200.10.10.1 255.255.255.252
R-DIST(config-if)#no shutdown

R-DIST(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R-DIST(config-if)#exit
R-DIST(config)#interface serial0/0/1
R-DIST(config-if)#description LINK-R1-DATA-CENTER
R-DIST(config-if)#ip address 10.0.0.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/0
R-DIST(config-if)#description LINK-R2-DATA-CENTER
R-DIST(config-if)#ip address 10.0.0.5 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/1
R-DIST(config-if)#description LINK-FILIAL-A-R3-OSPF
R-DIST(config-if)#ip address 10.0.10.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/0
R-DIST(config-if)#description LINK-FILIAL-B-R4-EIGRP
R-DIST(config-if)#ip address 10.0.20.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/1
R-DIST(config-if)#description LINK-FILIAL-C-R5-RIP
R-DIST(config-if)#ip address 10.0.30.1 255.255.255.252
R-DIST(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/1, changed state to down
R-DIST(config-if)#end
R-DIST#
%SYS-5-CONFIG_I: Configured from console by console

R-DIST#vr
Building configuration...
[OK]
R-DIST#
```

R1:

```
R1(config)#
R1(config)#
R1(config)#interface serial0/0/0
R1(config-if)#description UPLINK-R-DIST
R1(config-if)#ip address 10.0.0.2 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1(config-if)#interface giga0/1
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#interface giga0/1.10
R1(config-subif)#
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

R1(config-subif)#description VLAN10-ADMINISTRACAO
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config-subif)#interface giga0/1.20
R1(config-subif)#
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.20, changed state to up

R1(config-subif)#description VLAN20-VENDAS
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.192
R1(config-subif)#exit
R1(config-subif)#interface giga0/1.30
R1(config-subif)#
R1(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up

R1(config-subif)#description VLAN30-FINANCEIRO
R1(config-subif)#encapsulation dot1Q 30
R1(config-subif)#ip address 192.168.30.1 255.255.255.192
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#vr
Building configuration...
```

R-DIST: Configuração dos endereços IP nas interfaces WAN (ISP, R1, R2) e nos links para as filiais R3, R4 e R5.

R2:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial0/0/0
Router(config-if)#exit
Router(config)#hostname R3
R3(config)#
R3(config)#
R3(config)#
R3(config)#interface serial0/0/0
R3(config-if)#description WAN-LINK-R-DIST
R3(config-if)#ip address 10.0.10.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
R3(config-if)#interface giga0/0
R3(config-if)#description LAN-FILIAL-A-OSPF
R3(config-if)#ip address 10.0.1.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R3(config-if)#end
R3#
$SYS-5-CONFIG_I: Configured from console by console

R3#vr
Building configuration...
[OK]
R3#
```

R3:

R4:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#
R4(config)#interface serial0/0/0
R4(config-if)#description WAN-LINK-R-DIST
R4(config-if)#ip address 10.0.20.2 255.255.255.252
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
R4(config-if)#interface giga0/0
R4(config-if)#description LAN-FILIAL-B-RIGRP
R4(config-if)#ip address 10.0.2.1 255.255.255.0
R4(config-if)#no shutdown

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R4(config-if)#end
R4#
$SYS-5-CONFIG_I: Configured from console by console

R4#vr
Building configuration...
[OK]
R4#
```

R5:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R5
R5(config)#
R5(config)#interface serial0/0/0
R5(config-if)#description WAN-LINK-R-DIST
R5(config-if)#ip address 10.0.30.2 255.255.255.252
R5(config-if)#no shutdown

R5(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
R5(config-if)#interface giga0/0
R5(config-if)#description LAN-FILIAL-C-RIP
R5(config-if)#ip address 10.0.3.1 255.255.255.0
R5(config-if)#no shutdown

R5(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R5(config-if)#end
R5#
$SYS-5-CONFIG_I: Configured from console by console

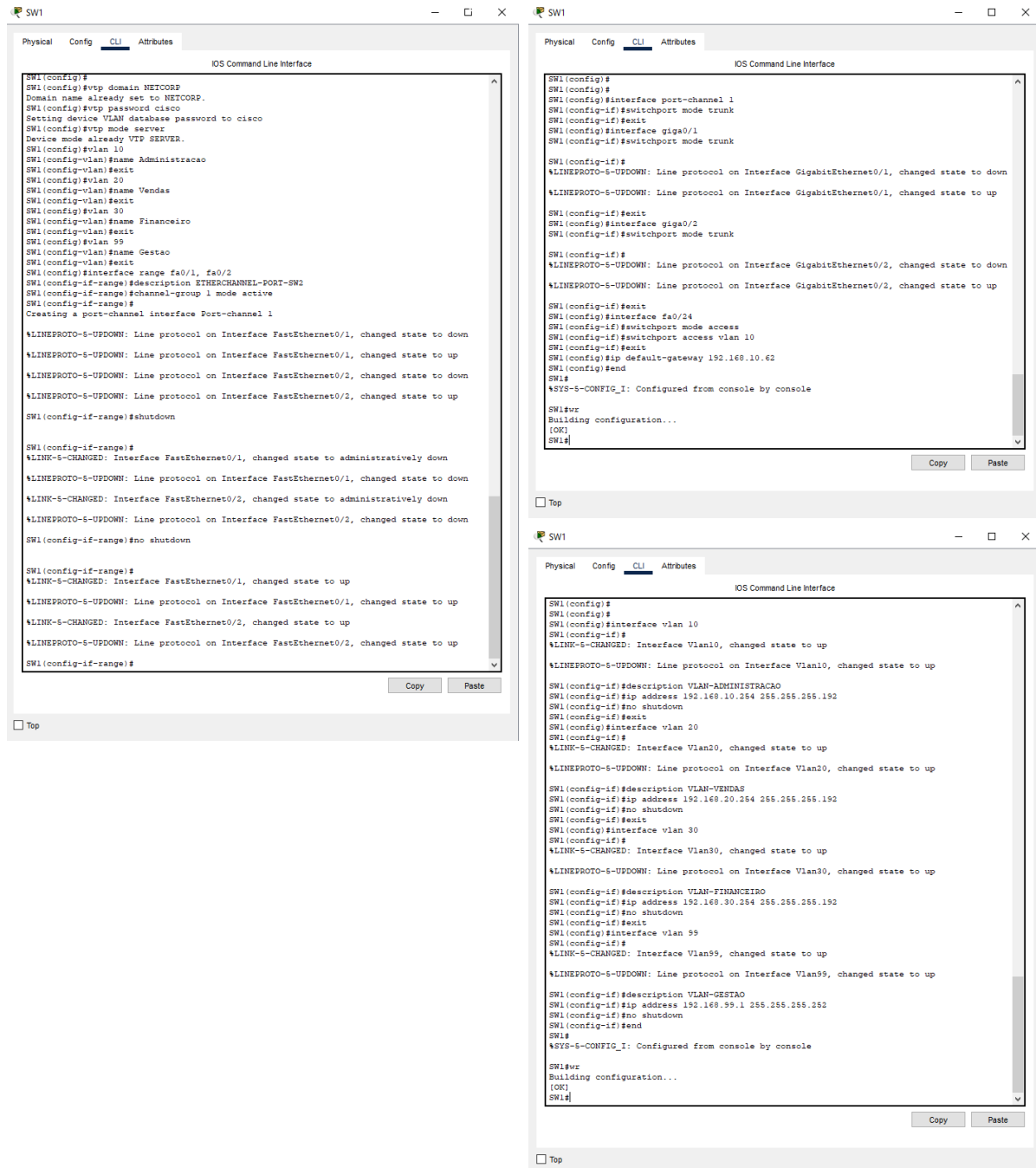
R5#vr
Building configuration...
[OK]
R5#
```

R1 e R2: Atribuição de IPs nas interfaces de ligação ao R-DIST e nas subinterfaces das VLANs 10, 20 e 30.

R3, R4, R5: Configuração dos endereços IP nas interfaces WAN (ligação ao R-DIST) e LAN (redes internas das filiais).

1) VLANs, VTP e Etherchannel

SW1:



```
SW1(config)#
SW1(config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW1(config)#vtp password cisco
Setting device VLAN database password to cisco
SW1(config)#vtp mode server
Device mode already VTP SERVER.
SW1(config)#vlan 10
SW1(config-vlan)#name Administracao
SW1(config-vlan)#exit
SW1(config-vlan)#vlan 20
SW1(config-vlan)#name Vendas
SW1(config-vlan)#exit
SW1(config-vlan)#vlan 30
SW1(config-vlan)#name Financeiro
SW1(config-vlan)#exit
SW1(config-vlan)#vlan 99
SW1(config-vlan)#name Gestao
SW1(config-vlan)#exit
SW1(config)#interface range fa0/1, fa0/2
SW1(config-if-range)#description ETHERCHANNEL-PORT-SW2
SW1(config-if-range)#channel-group 1 mode active
SW1(config-if-range)#
Creating a port-channel interface Port-channel 1
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
SW1(config-if-range)#shutdown
SW1(config-if-range)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-6-CHANGED: Interface FastEthernet0/2, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
SW1(config-if-range)#no shutdown
SW1(config-if-range)#
%LINK-6-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-6-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
SW1(config-if-range)#

SW1(config)#
SW1(config)#
SW1(config)#interface port-channel 1
SW1(config-if)#switchport mode trunk
SW1(config-if)#exit
SW1(config)#interface giga0/1
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW1(config-if)#exit
SW1(config)#interface giga0/2
SW1(config-if)#switchport mode trunk

SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
SW1(config-if)#exit
SW1(config)#interface fa0/24
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#exit
SW1(config)#ip default-gateway 192.168.10.62
SW1(config)#end
%SYS-5-CONFIG_I: Configured from console by console

SW1#
Building configuration...
[OK]
SW1#

SW1(config)#
SW1(config)#
SW1(config)#interface vlan 10
SW1(config-if)#
%LINK-6-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

SW1(config-if)#description VLAN-ADMINISTRACAO
SW1(config-if)#ip address 192.168.10.254 255.255.255.192
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#interface vlan 20
SW1(config-if)#
%LINK-6-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW1(config-if)#description VLAN-VENDAS
SW1(config-if)#ip address 192.168.20.254 255.255.255.192
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#interface vlan 30
SW1(config-if)#
%LINK-6-CHANGED: Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

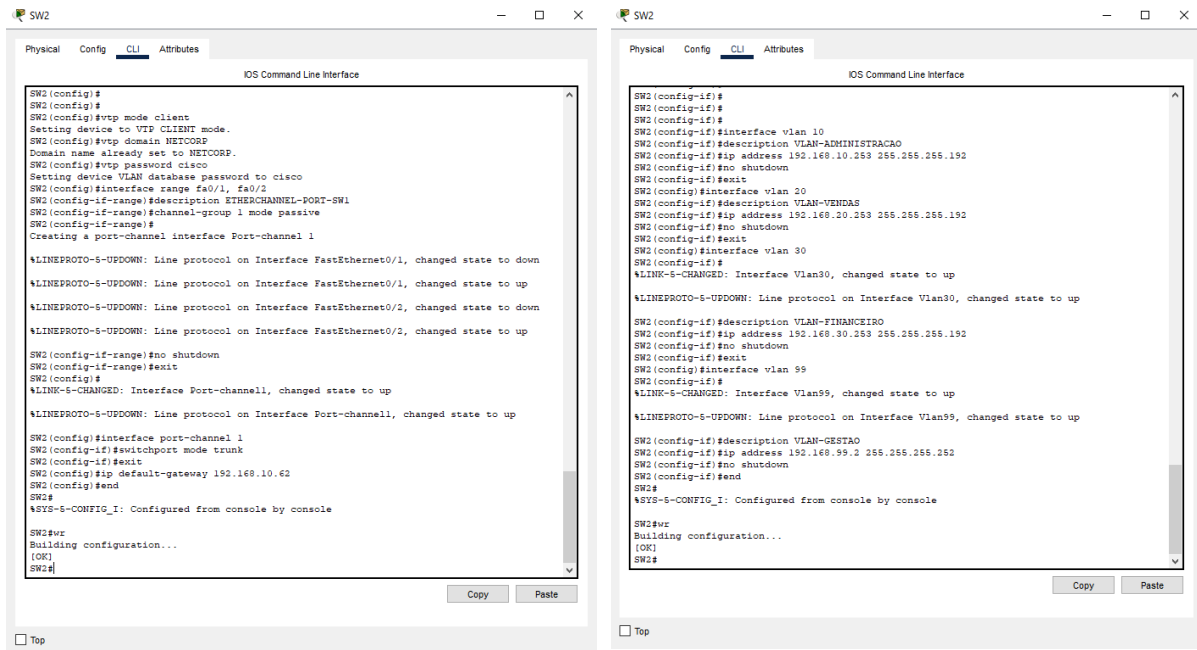
SW1(config-if)#description VLAN-FINANCEIRO
SW1(config-if)#ip address 192.168.30.254 255.255.255.192
SW1(config-if)#no shutdown
SW1(config-if)#exit
SW1(config)#interface vlan 99
SW1(config-if)#
%LINK-6-CHANGED: Interface Vlan99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up

SW1(config-if)#description VLAN-GESTAO
SW1(config-if)#ip address 192.168.99.1 255.255.255.252
SW1(config-if)#no shutdown
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#
Building configuration...
[OK]
SW1#
```

SW1 (VTP Server): Criação das VLANs 10, 20, 30 e 99. Ativação do VTP Server e configuração do EtherChannel com LACP para SW2.

SW2:



```
SW2 (config)#
SW2 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW2 (config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW2 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW2 (config)#interface range fa0/1, fa0/2
SW2 (config-if-range)#description ETHERCHANNEL-PORT-SW1
SW2 (config-if-range)#channel-group 1 mode passive
SW2 (config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW2 (config-if-range)#no shutdown
SW2 (config-if-range)#exit
SW2 (config)#
%LINK-6-CHANGED: Interface Port-channel1, changed state to up

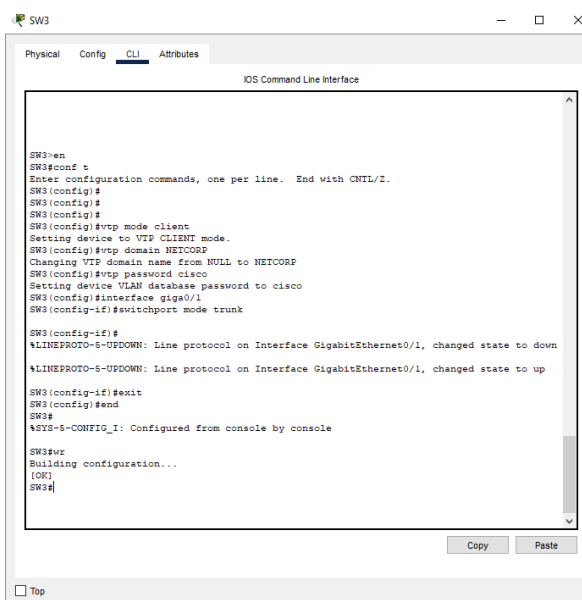
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW2 (config)#interface port-channel 1
SW2 (config-if)#switchport mode trunk
SW2 (config-if)#exit
SW2 (config)#ip default-gateway 192.168.10.62
SW2 (config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#vr
Building configuration...
[OK]
SW2#
```

SW2 (VTP Client): Configuração como VTP Client e participação no EtherChannel com SW1.

SW3:



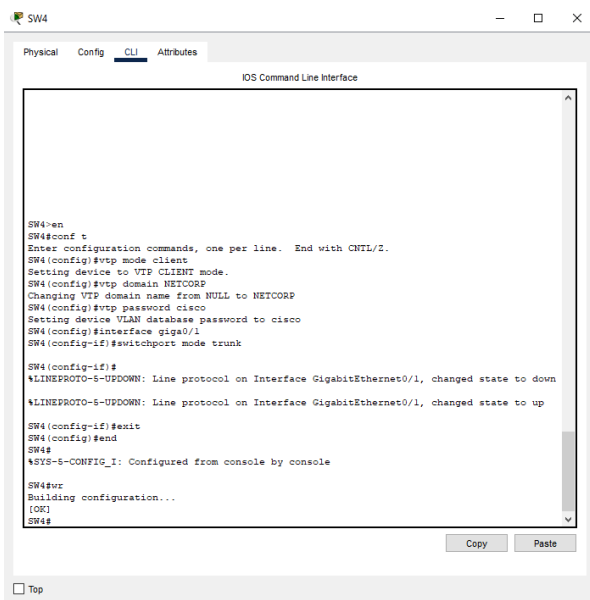
```
SW3>en
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3 (config)#
SW3 (config)#
SW3 (config)#
SW3 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW3 (config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW3 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW3 (config)#interface gig0/1
SW3 (config-if)#switchport mode trunk

SW3 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW3 (config-if)#exit
SW3 (config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#vr
Building configuration...
[OK]
SW3#
```

SW4:



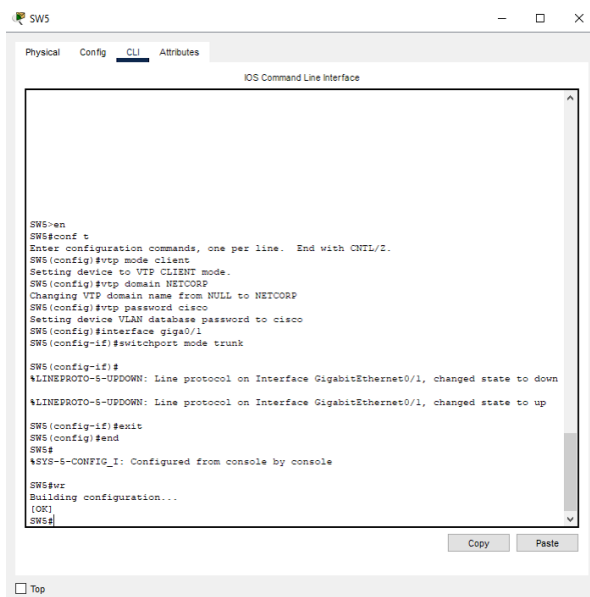
```
SW4>en
SW4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW4 (config)#vtp mode client
Setting device to VTP CLIENT mode.
SW4 (config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW4 (config)#vtp password cisco
Setting device VLAN database password to cisco
SW4 (config)#interface gig0/1
SW4 (config-if)#switchport mode trunk

SW4 (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW4 (config-if)#exit
SW4 (config)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console

SW4#vr
Building configuration...
[OK]
SW4#
```

SW5:



```
SW5>en
SW5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW5(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW5(config)#vtp domain NETCORP
Changing VTP domain name from NULL to NETCORP
SW5(config)#vtp password cisco
Setting device VLAN database password to cisco
SW5(config)#interface giga0/1
SW5(config-if)#switchport mode trunk

SW5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
SW5(config-if)#exit
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console

SW5#wr
Building configuration...
[OK]
SW5#
```

SW3, SW4, SW5: Configuração como VTP Clients e ativação das portas de trunk para os routers das filiais.

2) Redundância com HSRP

R1:

```
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface giga0/1.10
R1(config-subif)#standby 10 ip 192.168.10.62
R1(config-subif)#standby 10 priority 110
R1(config-subif)#standby 10 preempt
R1(config-subif)#standby 10 authentication md5 key-string cisco

% Invalid input detected at '^' marker.

R1(config-subif)#
%HSRP-6-STATECHANGE: GigabitEthernet0/1.10 Grp 10 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.10 Grp 10 state Standby -> Active

R1(config-subif)#interface giga0/1.20
R1(config-subif)#standby 20 ip 192.168.20.62
R1(config-subif)#standby 20 priority 110
R1(config-subif)#standby 20 preempt
R1(config-subif)#interface giga0/1.30

%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Standby -> Active

R1(config-subif)#standby 30 ip 192.168.30.62
R1(config-subif)#standby 30 priority 110
R1(config-subif)#standby 30 preempt
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#
```

R2:

```
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface giga0/2.10
R2(config-subif)#standby 10 ip 192.168.10.62
R2(config-subif)#standby 10 priority 100
R2(config-subif)#
%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Speak -> Standby

R2(config-subif)#interface giga0/2.20
R2(config-subif)#standby 20 ip 192.168.20.62
R2(config-subif)#standby 20 priority 100
R2(config-subif)#interface giga0/2.30
R2(config-subif)#
%HSRP-6-STATECHANGE: GigabitEthernet0/2.20 Grp 20 state Speak -> Standby

R2(config-subif)#standby 30 ip 192.168.30.62
R2(config-subif)#standby 30 priority 100
R2(config-subif)#
%HSRP-6-STATECHANGE: GigabitEthernet0/2.30 Grp 30 state Speak -> Standby

R2(config-subif)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

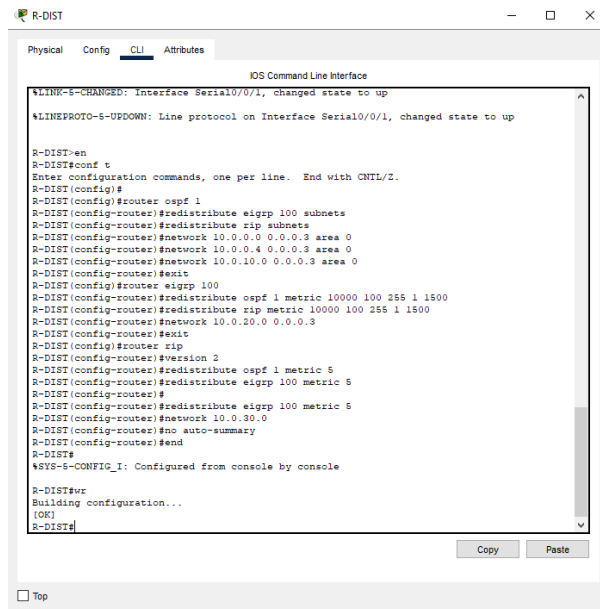
R2#wr
Building configuration...
[OK]
R2#
```

R1 (Ativo): Configuração do HSRP com prioridade 110 e preempt, tornando-o o gateway ativo para as VLANs 10, 20 e 30.

R2 (Standby): Configuração do HSRP com prioridade 100, garantindo redundância em caso de falha do R1.

3) Roteamento dinâmico e Redistribuição

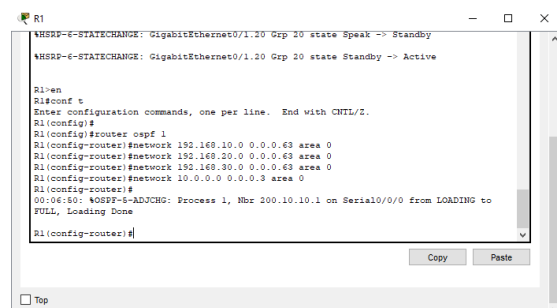
R-DIST:



```
R-DIST
Physical Config CLI Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
R-DIST>en
R-DIST#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#
R-DIST(config)#router ospf 1
R-DIST(config-router)#redistribute eigrp 100 subnets
R-DIST(config-router)#redistribute rip subnets
R-DIST(config-router)#network 10.0.0.0 0.0.0.3 area 0
R-DIST(config-router)#network 10.0.0.4 0.0.0.3 area 0
R-DIST(config-router)#network 10.0.10.0 0.0.0.3 area 0
R-DIST(config-router)#exit
R-DIST(config)#router eigrp 100
R-DIST(config-router)#redistribute ospf 1 metric 10000 100 255 1 1500
R-DIST(config-router)#redistribute rip metric 10000 100 255 1 1500
R-DIST(config-router)#network 10.0.20.0 0.0.0.3
R-DIST(config-router)#exit
R-DIST(config)#router rip
R-DIST(config-router)#version 2
R-DIST(config-router)#redistribute ospf 1 metric 5
R-DIST(config-router)#redistribute eigrp 100 metric 5
R-DIST(config-router)#
R-DIST(config-router)#redistribute eigrp 100 metric 5
R-DIST(config-router)#network 10.0.30.0
R-DIST(config-router)#no auto-summary
R-DIST(config-router)#end
R-DIST#
%SYS-5-CONFIG_I: Configured from console by console
R-DIST#vr
Building configuration...
(OX)
R-DIST#
```

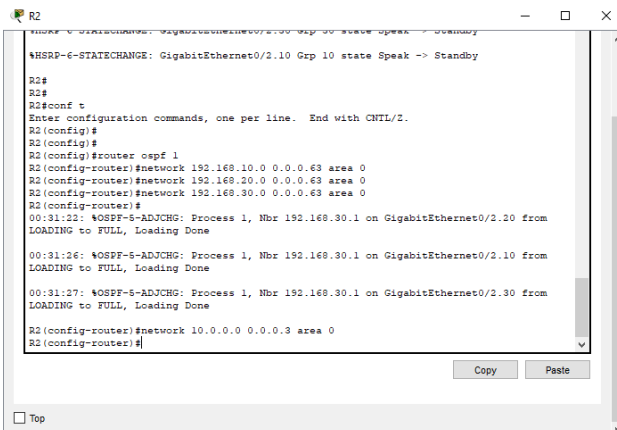
R-DIST: Redistribuição entre OSPF (Filial A), EIGRP (Filial B) e RIP (Filial C), garantindo conectividade total entre todas as redes.

R1:



```
R1
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: GigabitEthernet0/1.20 Grp 20 state Standby -> Active
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#router ospf 1
R1(config-router)#network 192.168.10.0 0.0.0.63 area 0
R1(config-router)#network 192.168.20.0 0.0.0.63 area 0
R1(config-router)#network 192.168.30.0 0.0.0.63 area 0
R1(config-router)#network 10.0.0.0 0.0.0.3 area 0
R1(config-router)#
00:06:50: %OSPF-5-ADJCHG: Process 1, Nbr 200.10.10.1 on Serial0/0/0 from LOADING to FULL, Loading Done
R1(config-router)#
```

R2:



```
R2
%HSRP-6-STATECHANGE: GigabitEthernet0/2.10 Grp 10 state Speak -> Standby
R2#
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#router ospf 1
R2(config-router)#network 192.168.10.0 0.0.0.63 area 0
R2(config-router)#network 192.168.20.0 0.0.0.63 area 0
R2(config-router)#network 192.168.30.0 0.0.0.63 area 0
R2(config-router)#
00:31:22: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.20 from LOADING to FULL, Loading Done
00:31:26: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.10 from LOADING to FULL, Loading Done
00:31:27: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on GigabitEthernet0/2.30 from LOADING to FULL, Loading Done
R2(config-router)#network 10.0.0.0 0.0.0.3 area 0
R2(config-router)#
```

R1 e R2: Ativação do OSPF para anunciar as VLANs do Data Center.

R3:

```
R3
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R3#
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.1.0 0.0.0.255 area 0
R3(config-router)#network 10.0.10.0 0.0.0.3 area 0
R3(config-router)#
00:32:41: %OSPF-5-ADJCHG: Process 1, Nbr 200.10.10.1 on Serial10/0/0 from LOADING to FULL, Loading Done
```

R4:

```
R4
$LINK-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R4#
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#
R4(config)#router eigrp 100
R4(config-router)#network 10.0.2.0 0.0.0.255
R4(config-router)#network 10.0.20.0 0.0.0.3
R4(config-router)#
%UAL-5-NBCHANGE: IP-EIGRP 100: Neighbor 10.0.20.1 (Serial10/0/0) is up: new adjacency
R4(config-router)#no auto-summary
R4(config-router)#
```

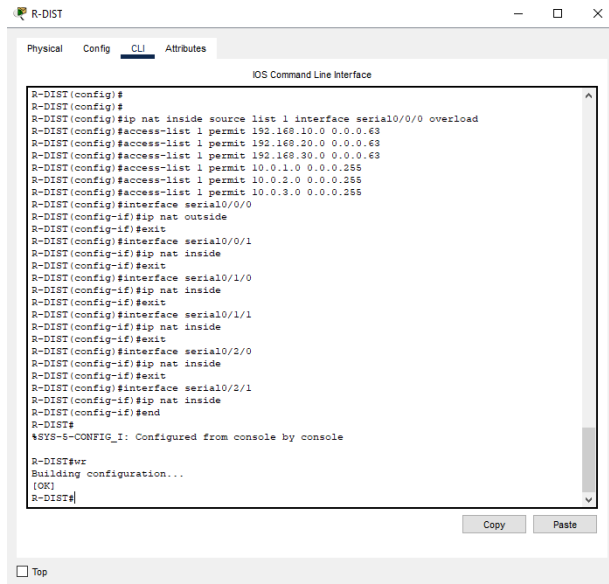
R5:

```
R5
$LINK-5-CHANGED: Interface Serial10/0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
$LINK-5-UPDOWN: Line protocol on Interface Serial10/0/0, changed state to up
R5#
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#network 10.0.3.0
R5(config-router)#network 10.0.30.0
R5(config-router)#no auto-summary
R5(config-router)#
```

R3, R4, R5: Ativação dos protocolos OSPF, EIGRP e RIP nas respetivas filiais.

4) NAT (PAT) em todos os routers

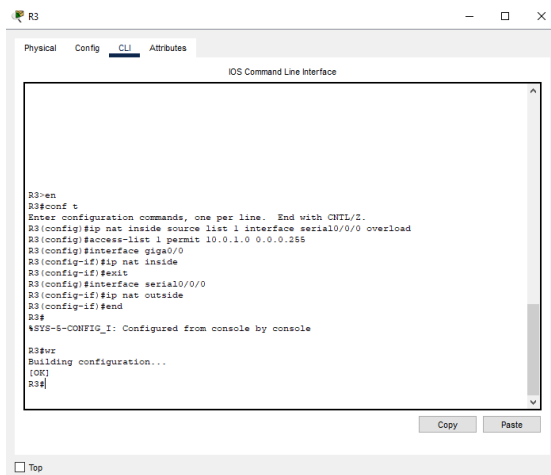
R-DIST:



```
R-DIST(config)#
R-DIST(config)#
R-DIST(config)#ip nat inside source list 1 interface serial0/0/0 overload
R-DIST(config)#access-list 1 permit 192.168.10.0 0.0.0.63
R-DIST(config)#access-list 1 permit 192.168.20.0 0.0.0.63
R-DIST(config)#access-list 1 permit 192.168.30.0 0.0.0.63
R-DIST(config)#access-list 1 permit 10.0.1.0 0.0.0.255
R-DIST(config)#access-list 1 permit 10.0.2.0 0.0.0.255
R-DIST(config)#access-list 1 permit 10.0.3.0 0.0.0.255
R-DIST(config)#interface serial0/0/0
R-DIST(config-if)#ip nat outside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/0/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/0
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/1/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/0
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#exit
R-DIST(config)#interface serial0/2/1
R-DIST(config-if)#ip nat inside
R-DIST(config-if)#end
R-DIST#
$SYS-S-CONFIG_I: Configured from console by console

R-DIST#vr
Building configuration...
[OK]
R-DIST#
```

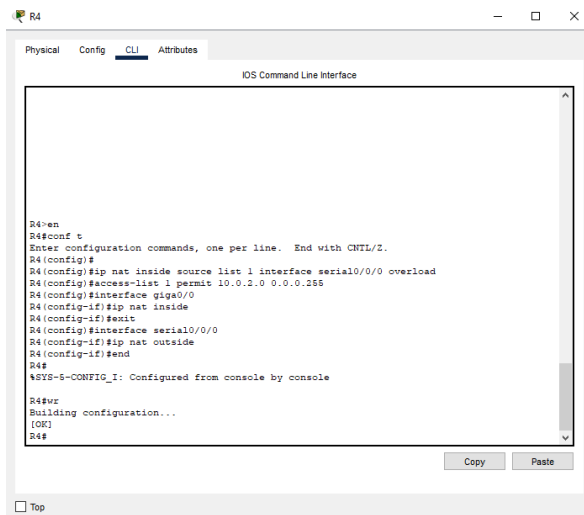
R3:



```
R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip nat inside source list 1 interface serial0/0/0 overload
R3(config)#access-list 1 permit 10.0.1.0 0.0.0.255
R3(config)#interface giga0/0
R3(config-if)#ip nat inside
R3(config-if)#exit
R3(config)#interface serial0/0/0
R3(config-if)#ip nat outside
R3(config-if)#end
R3#
$SYS-S-CONFIG_I: Configured from console by console

R3#vr
Building configuration...
[OK]
R3#
```

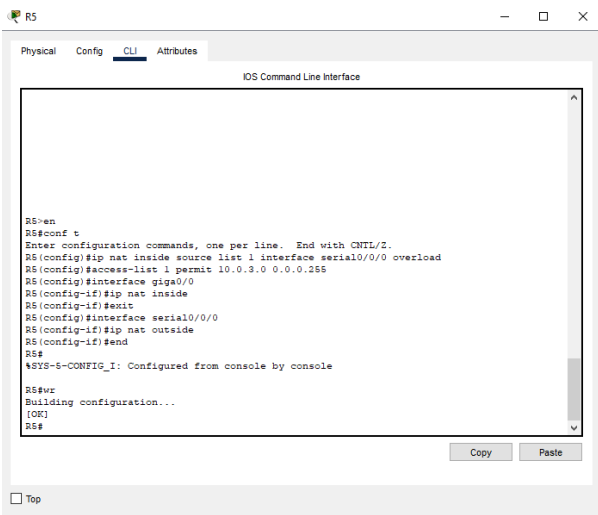
R4:



```
R4>en
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#ip nat inside source list 1 interface serial0/0/0 overload
R4(config)#access-list 1 permit 10.0.2.0 0.0.0.255
R4(config)#interface giga0/0
R4(config-if)#ip nat inside
R4(config-if)#exit
R4(config)#interface serial0/0/0
R4(config-if)#ip nat outside
R4(config-if)#end
R4#
$SYS-S-CONFIG_I: Configured from console by console

R4#vr
Building configuration...
[OK]
R4#
```

R5:



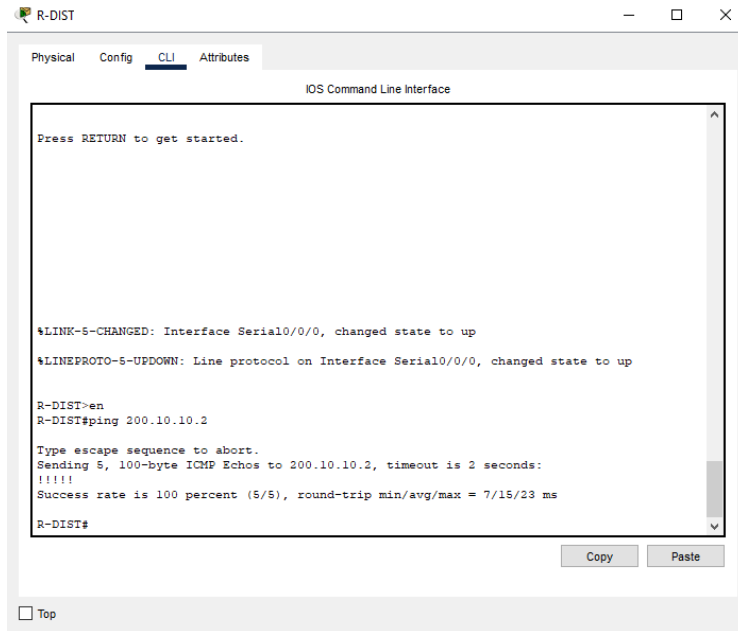
```
R5>en
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip nat inside source list 1 interface serial0/0/0 overload
R5(config)#access-list 1 permit 10.0.3.0 0.0.0.255
R5(config)#interface giga0/0
R5(config-if)#ip nat inside
R5(config-if)#exit
R5(config)#interface serial0/0/0
R5(config-if)#ip nat outside
R5(config-if)#end
R5#
$SYS-S-CONFIG_I: Configured from console by console

R5#vr
Building configuration...
[OK]
R5#
```

R-DIST: Configuração do NAT overload com ACL 1 a definir as redes internas autorizadas.

R3, R4, R5: Configuração de NAT local para permitir acesso à Internet a partir das redes das filiais.

Agora testo um ping do router R-DIST para o endereço IP do ISP (200.10.10.2), obtendo uma resposta com sucesso.



```
R-DIST
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R-DIST>en
R-DIST#ping 200.10.10.2

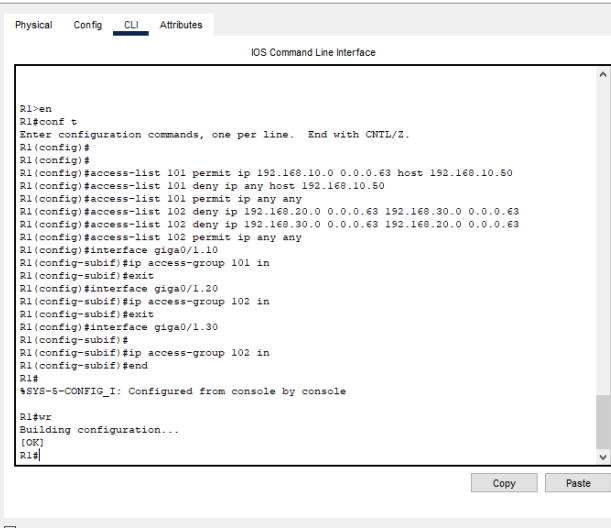
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/15/23 ms

R-DIST#
```

A comunicação dos PCs será testada mais tarde, após a configuração do serviço DHCP.

5) ACLs – Listas de Controlo de Acesso

R1:

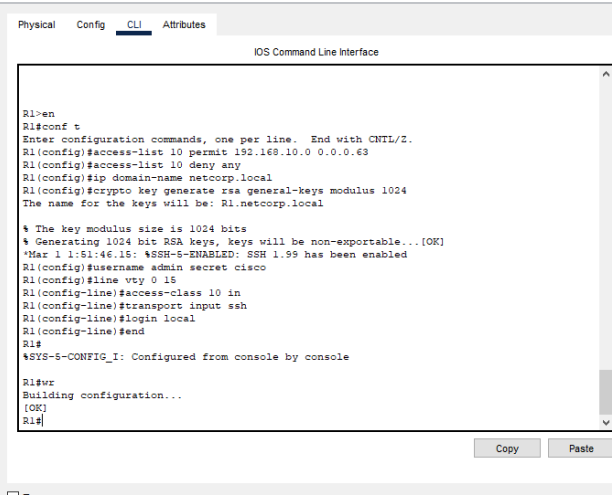


```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#access-list 101 permit ip 192.168.10.0 0.0.0.63 host 192.168.10.50
R1(config)#access-list 101 deny ip any host 192.168.10.50
R1(config)#access-list 101 permit ip any any
R1(config)#access-list 102 deny ip 192.168.20.0 0.0.0.63 192.168.30.0 0.0.0.63
R1(config)#access-list 102 deny ip 192.168.30.0 0.0.0.63 192.168.20.0 0.0.0.63
R1(config)#access-list 102 permit ip any any
R1(config)#interface giga0/1.10
R1(config-subif)#ip access-group 101 in
R1(config-subif)#exit
R1(config)#interface giga0/1.20
R1(config-subif)#ip access-group 102 in
R1(config-subif)#exit
R1(config)#interface giga0/1.30
R1(config-subif)#ip access-group 102 in
R1(config-subif)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#
```

R1: Aplicação de ACLs para restringir acesso ao SRV1 (apenas VLAN 10) e bloquear comunicação entre VLAN 20 e VLAN 30.

Todos os dispositivos:



```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R1(config)#access-list 10 deny any
R1(config)#ip domain-name netcorp.local
R1(config)#crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R1.netcorp.local

% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 1:51:46.18: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#username admin secret cisco
R1(config)#line vty 0 15
R1(config-line)#access-class 10 in
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

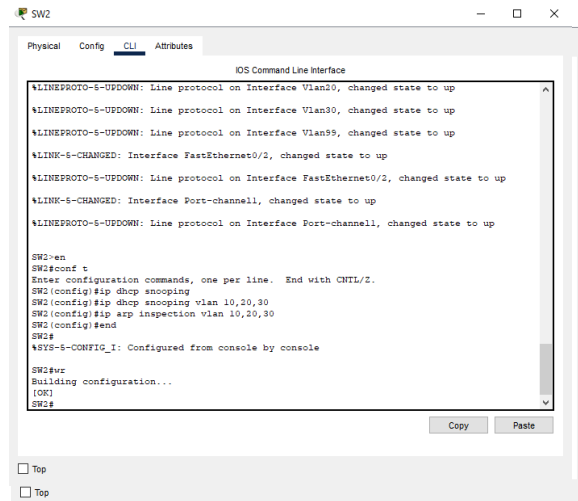
R1#wr
Building configuration...
[OK]
R1#
```

Todos os dispositivos: Criação da ACL 10 para restringir SSH à VLAN 10 e ativação do SSH com desativação do Telnet.

6) Segurança de Camada 2

SW1:

SW2:



```
SW2
Physical Config CLI Attributes
IOS Command Line Interface

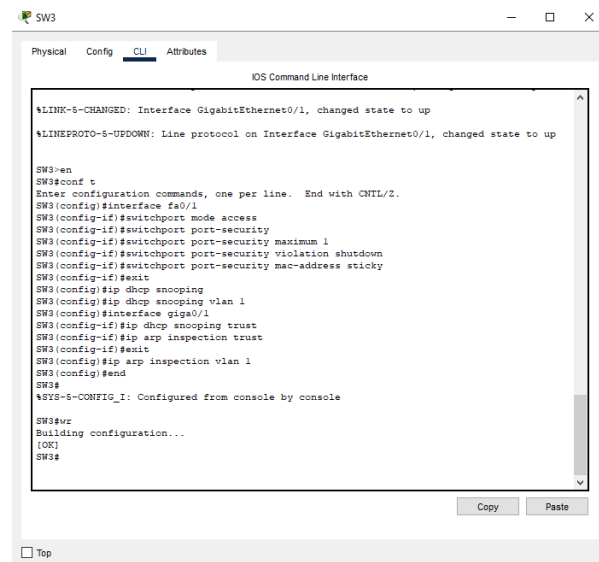
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan39, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW2>en
SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#ip dhcp snooping
SW2(config)#ip dhcp snooping vlan 10,20,30
SW2(config)#ip arp inspection vlan 10,20,30
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#wr
Building configuration...
[OK]
SW2#
```

SW1 e SW2: Ativação do DHCP Snooping e DAI, com definição das portas trusted (R1, R2, SRV1).

SW3, SW4 e SW5:



```
SW3
Physical Config CLI Attributes
IOS Command Line Interface

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW3>en
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#interface fa0/1
SW3(config-if)#switchport mode access
SW3(config-if)#switchport port-security
SW3(config-if)#switchport port-security maximum 1
SW3(config-if)#switchport port-security violation shutdown
SW3(config-if)#switchport port-security mac-address sticky
SW3(config-if)#exit
SW3(config)#ip dhcp snooping
SW3(config)#ip dhcp snooping vlan 1
SW3(config)#interface giga0/1
SW3(config-if)#ip dhcp snooping trust
SW3(config-if)#ip arp inspection trust
SW3(config-if)#exit
SW3(config)#ip arp inspection vlan 1
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

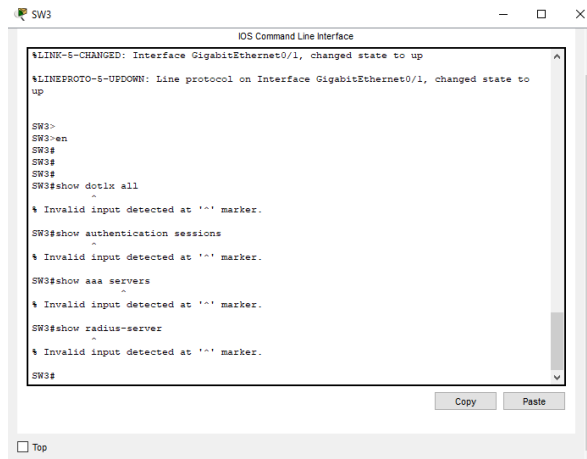
SW3#wr
Building configuration...
[OK]
SW3#
```

Configuração igual para os 3, alterando apenas as interfaces usadas caso sejam diferentes.

SW3, SW4, SW5: Configuração de Port-Security nas portas dos PCs e ativação do DHCP Snooping e DAI nas VLANs das filiais.

7) Autenticação AAA, IEEE 802.1X e SSH

Durante a implementação do Exercício 7, foram realizadas várias tentativas de configurar o 802.1X com autenticação RADIUS, seguindo os procedimentos utilizados nas aulas, no entanto, mesmo após configurar tudo, os comandos de verificação fundamentais não estavam disponíveis:



```
SW3
IOS Command Line Interface

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW3>
SW3#
SW3#
SW3#
SW3#show dot1x all
% Invalid input detected at '^' marker.

SW3#show authentication sessions
% Invalid input detected at '^' marker.

SW3#show aaa servers
% Invalid input detected at '^' marker.

SW3#show radius-server
% Invalid input detected at '^' marker.

SW3#
```

A porta permanece bloqueada e o DHCP falha, mesmo com as credenciais corretas no PC.

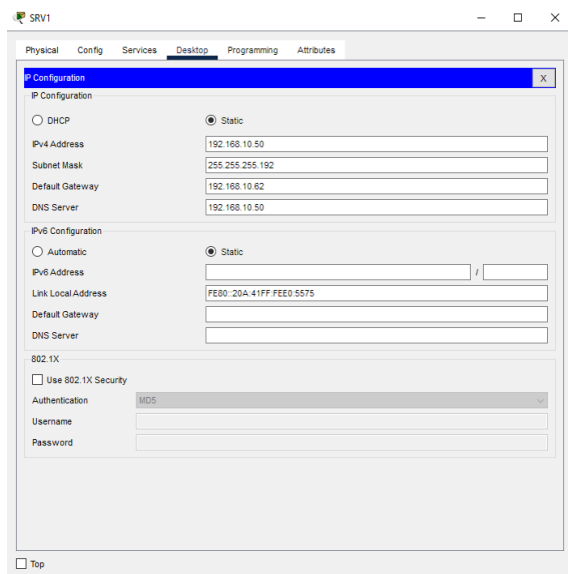
Não foi possível determinar com certeza se esta falha resulta de uma limitação do simulador (versão 8.2.2.0400) ou de algum erro na configuração da minha parte, apesar de aparentemente correto, não é aceite pelo Packet Tracer.

O que é certo é que, nesta versão, ambos o 802.1X e RADIUS não se comportam como esperado e não foi possível torná-los funcionais.

Por causa disto decidi fazer este exercício sem utilizar 802.1X ou RADIUS, para cumprir o objetivo do enunciado foi implementada uma solução alternativa funcional, composta por:

- Port-Security com sticky MAC para controlo de acesso por porta.
- Autenticação local com os utilizadores user1/cinel e admin/cisco.
- SSH ativo em todos os equipamentos com ACL restrita à VLAN 10.
- DHCP relay (ip helper-address) para atribuição dinâmica de IPs.

SRV1:



SRV1

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

Pv4 Address: 192.168.10.50

Subnet Mask: 255.255.255.192

Default Gateway: 192.168.10.62

DNS Server: 192.168.10.50

IPv6 Configuration

☐ Automatic ☒ Static

Pv6 Address: /

Link Local Address: FE80::20A:41FF:FE00:5575

Default Gateway:

DNS Server:

802.1X

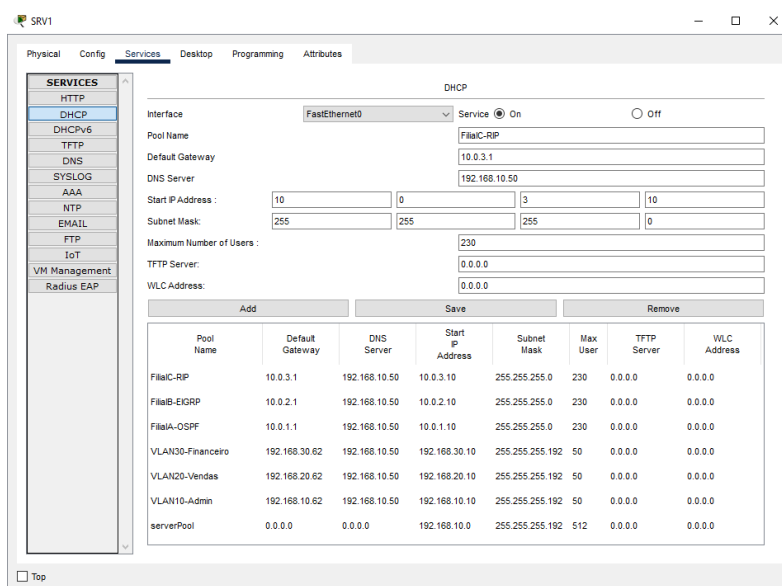
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top



SRV1

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: FiliaC-RP

Default Gateway: 10.0.3.1

DNS Server: 192.168.10.50

Start IP Address: 10 0 3 10

Subnet Mask: 255 255 255 0

Maximum Number of Users: 230

TFTP Server: 0.0.0.0

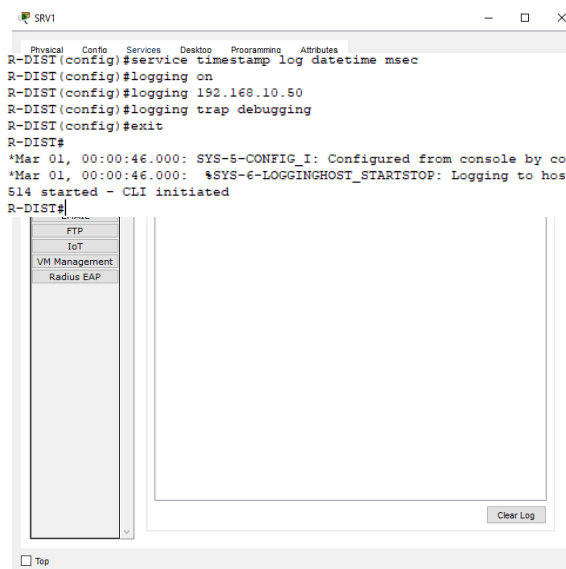
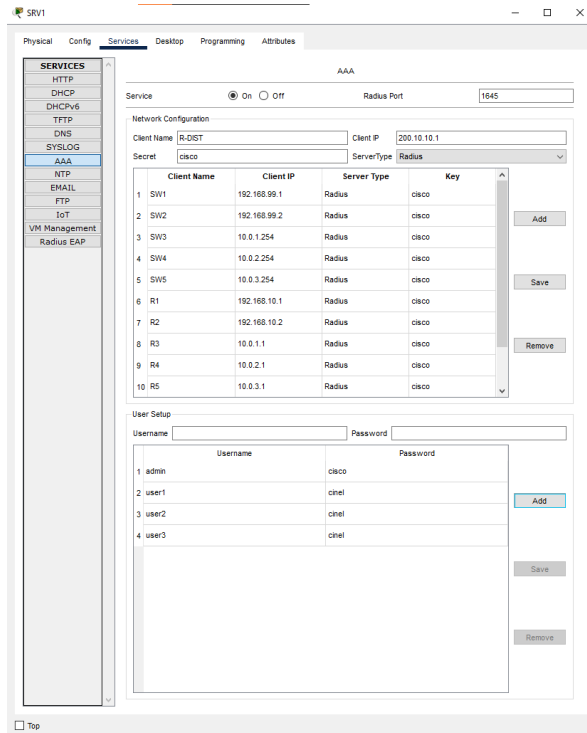
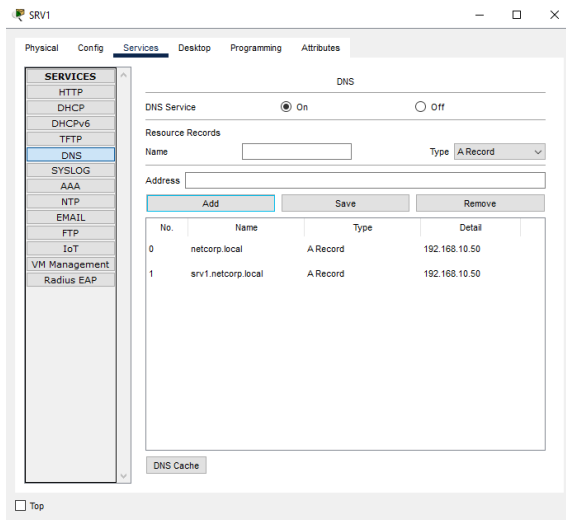
WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
FiliaC-RP	10.0.3.1	192.168.10.50	10.0.3.10	255.255.255.0	230	0.0.0.0	0.0.0.0
FiliaB-EIGRP	10.0.2.1	192.168.10.50	10.0.2.10	255.255.255.0	230	0.0.0.0	0.0.0.0
FiliaA-OSPF	10.0.1.1	192.168.10.50	10.0.1.10	255.255.255.0	230	0.0.0.0	0.0.0.0
VLAN30-Financeiro	192.168.30.62	192.168.10.50	192.168.30.10	255.255.255.192	50	0.0.0.0	0.0.0.0
VLAN20-Vendas	192.168.20.62	192.168.10.50	192.168.20.10	255.255.255.192	50	0.0.0.0	0.0.0.0
VLAN10-Admin	192.168.10.62	192.168.10.50	192.168.10.10	255.255.255.192	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.0	255.255.255.192	512	0.0.0.0	0.0.0.0

☐ Top

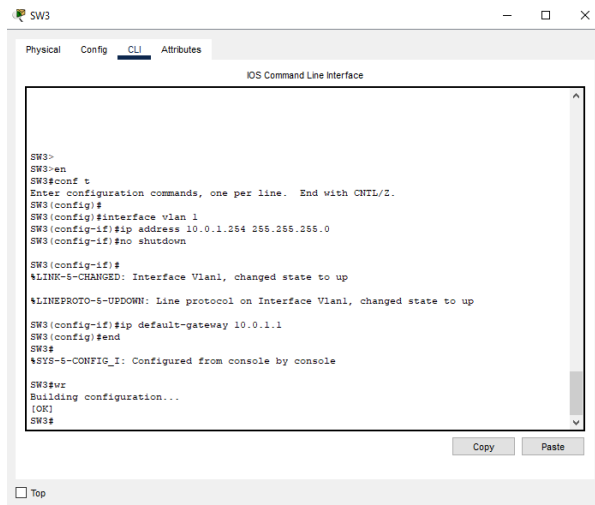
Apesar de estar ON acabei por não usar AAA:



R-DIST:

SRV1: Configuração do servidor com IP fixo e ativação dos serviços DHCP e DNS para todas as redes.

SW3:

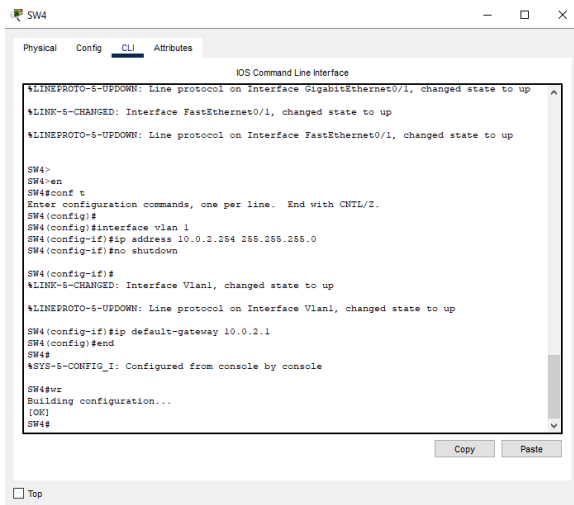


```
SW3>
SW3#en
SW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#
SW3(config)#interface vlan 1
SW3(config-if)#ip address 10.0.1.254 255.255.255.0
SW3(config-if)#no shutdown

SW3(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW3(config-if)#ip default-gateway 10.0.1.1
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#vr
Building configuration...
(OX)
SW3#
```

SW4:

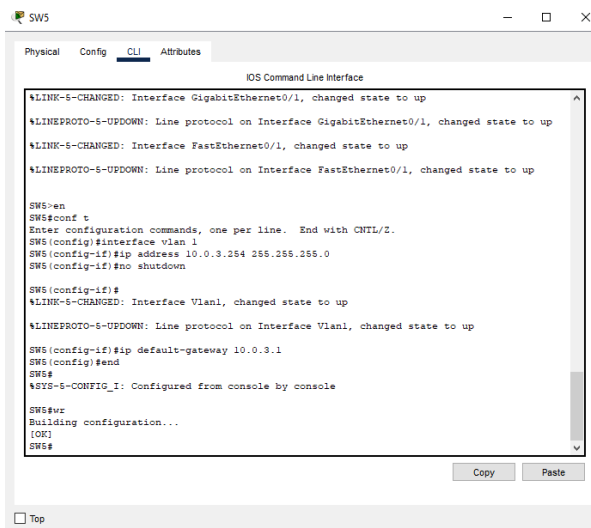


```
SW4>
SW4#en
SW4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW4(config)#
SW4(config)#interface vlan 1
SW4(config-if)#ip address 10.0.2.254 255.255.255.0
SW4(config-if)#no shutdown

SW4(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW4(config-if)#ip default-gateway 10.0.2.1
SW4(config)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console

SW4#vr
Building configuration...
(OX)
SW4#
```

SW5:



```
SW5>
SW5#en
SW5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW5(config)#interface vlan 1
SW5(config-if)#ip address 10.0.3.254 255.255.255.0
SW5(config-if)#no shutdown

SW5(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
SW5(config-if)#ip default-gateway 10.0.3.1
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console

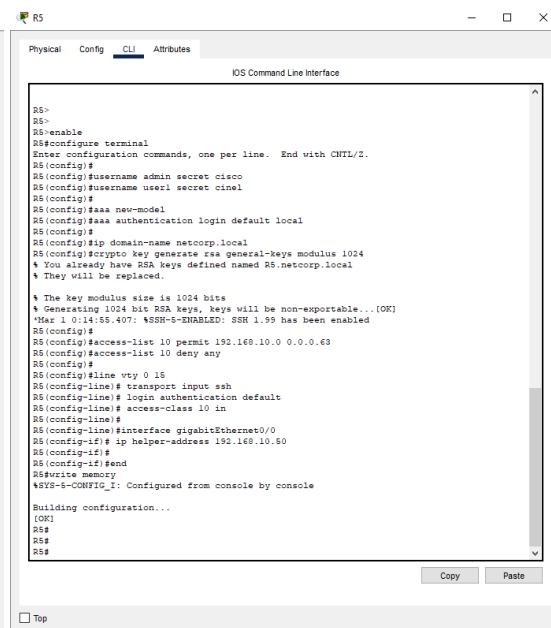
SW5#vr
Building configuration...
(OX)
SW5#
```

SW3, SW4, SW5:



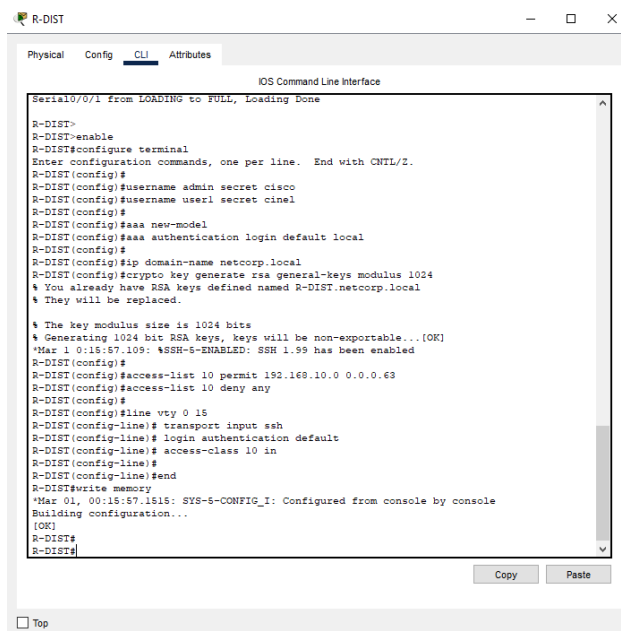
```
SW5#
SW5#
SW5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW5(config)#vtp mode client
Device mode already VTP CLIENT.
SW5(config)#vtp domain NETCORP
Domain name already set to NETCORP.
SW5(config)#vtp password cisco
Password already set to cisco
SW5(config)#
SW5(config)#interface gigabitEthernet0/1
SW5(config-if)# switchport mode trunk
SW5(config-if)# description TRUNK-RS
SW5(config-if)#
SW5(config-if)#username admin secret cisco
SW5(config-if)#username user1 secret cinel
SW5(config-if)#username user2 secret cinel
SW5(config-if)#username user3 secret cinel
SW5(config-if)#
SW5(config)#aaa new-model
SW5(config)#aaa authentication login default local
SW5(config)#
SW5(config)#ip domain-name netcorp.local
SW5(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named SW5.netcorp.local
% They will be replaced.
% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:12:53.897: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW5(config)#
SW5(config)#access-list 10 permit 192.168.10.0 0.0.0.63
SW5(config)#access-list 10 deny any
SW5(config)#
SW5(config)#line vty 0 15
SW5(config-line)# transport input ssh
SW5(config-line)# login authentication default
SW5(config-line)# access-class 10 in
SW5(config-line)#
SW5(config-line)#interface fa0/1
SW5(config-if)# switchport mode access
SW5(config-if)# switchport port-security
SW5(config-if)# switchport port-security maximum 1
SW5(config-if)# switchport port-security violation shutdown
SW5(config-if)# switchport port-security mac-address sticky
SW5(config-if)# no shutdown
SW5(config-if)#
SW5(config-if)#end
SW5#write memory
*SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
SW5#
SW5#
```

R3, R4, R5:



```
R5>
R5>
R5>enable
R5#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#
R5(config)#username admin secret cisco
R5(config)#username user1 secret cinel
R5(config)#
R5(config)#aaa new-model
R5(config)#aaa authentication login default local
R5(config)#
R5(config)#ip domain-name netcorp.local
R5(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named R5.netcorp.local
% They will be replaced.
% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:14:55.407: %SSH-5-ENABLED: SSH 1.99 has been enabled
R5(config)#
R5(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R5(config)#access-list 10 deny any
R5(config)#
R5(config)#line vty 0 15
R5(config-line)# transport input ssh
R5(config-line)# login authentication default
R5(config-line)# access-class 10 in
R5(config-line)#
R5(config-line)#interface gigabitEthernet0/0
R5(config-if)# ip helper-address 192.168.10.50
R5(config-if)#
R5(config-if)#end
R5#write memory
*SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
R5#
R5#
R5#
```

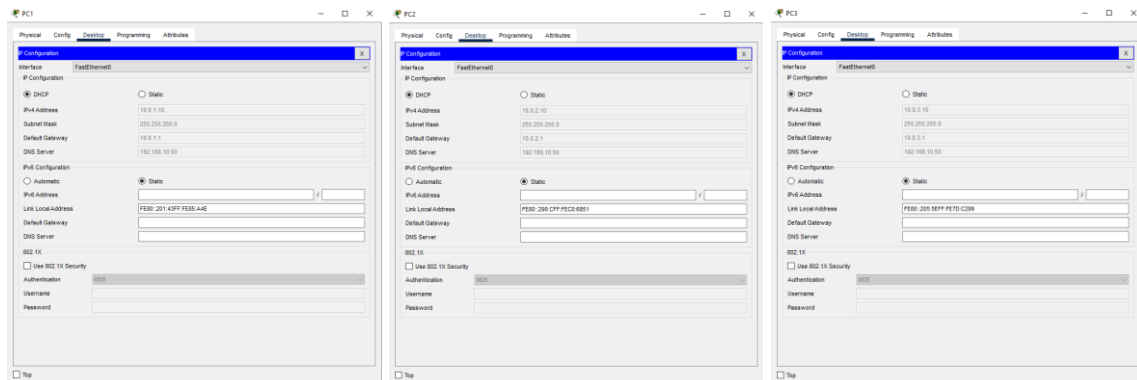
R-DIST, R1, R2:



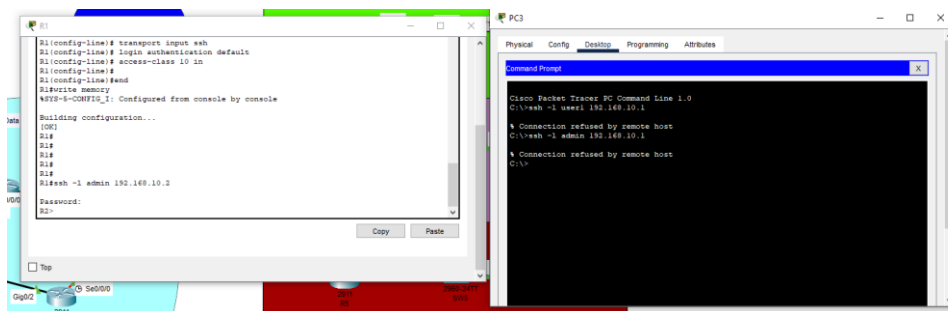
```
R-DIST
R-DIST
R-DIST>
R-DIST>enable
R-DIST#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#
R-DIST(config)#username admin secret cisco
R-DIST(config)#username user1 secret cinel
R-DIST(config)#
R-DIST(config)#aaa new-model
R-DIST(config)#aaa authentication login default local
R-DIST(config)#
R-DIST(config)#ip domain-name netcorp.local
R-DIST(config)#crypto key generate rsa general-keys modulus 1024
% You already have RSA keys defined named R-DIST.netcorp.local
% They will be replaced.
% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
*Mar 1 0:15:57.109: %SSH-5-ENABLED: SSH 1.99 has been enabled
R-DIST(config)#
R-DIST(config)#access-list 10 permit 192.168.10.0 0.0.0.63
R-DIST(config)#access-list 10 deny any
R-DIST(config)#
R-DIST(config)#line vty 0 15
R-DIST(config-line)# transport input ssh
R-DIST(config-line)# login authentication default
R-DIST(config-line)# access-class 10 in
R-DIST(config-line)#
R-DIST(config-line)#end
R-DIST#write memory
*Mar 01, 00:15:57.161S: SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
R-DIST#
R-DIST#
```

Switches (SW3, SW4, SW5): Criação de utilizadores locais, ativação do AAA com autenticação local, configuração do SSH com ACL e implementação de Port-Security.

Routers: Configuração de autenticação local e SSH em todos os routers, com ip helper-address nos routers das filiais para DHCP relay.

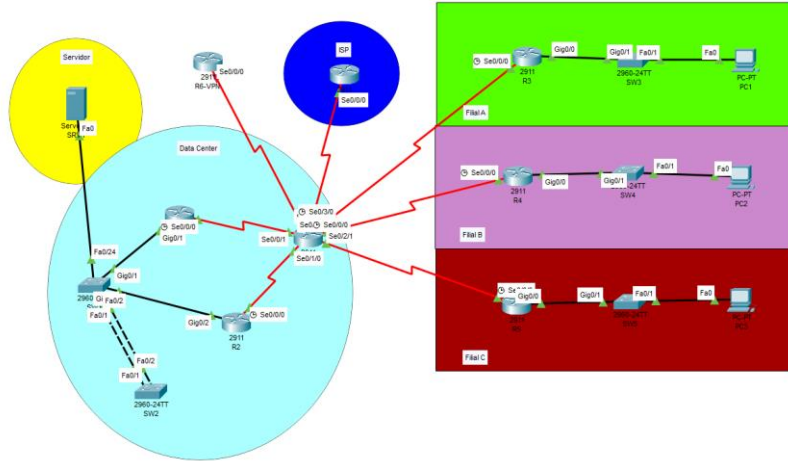


Apenas dispositivos na vlan 10 conseguem fazer ssh, acl bloqueia o resto.



8) VPN – Virtual Private Network

Para este exercício vou adicionar um router com uma rede diferente e conecta-lo ao R-DIST: R6-VPN (se0/0/0) → (se0/3/0) R-DIST



Antes de configurar o R-DIST primeiro adiciono o securityk9:

R-DIST# license boot module c2900 technology-package securityk9
R-DIST# reload

R-DIST:

```

R-DIST#
R-DIST#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-DIST(config)#interface Serial0/3/0
R-DIST(config-if)#description LINK-R6
R-DIST(config-if)#ip address 200.10.20.1 255.255.255.252
R-DIST(config-if)#clock rate 64000
This command applies only to DCE interfaces
R-DIST(config-if)#no shutdown
R-DIST(config-if)#exit
R-DIST(config)#interface Tunnel0
R-DIST(config-if)#
R-DIST(config-if)#ip address 10.0.99.1 255.255.255.252
R-DIST(config-if)#tunnel source Serial0/3/0
R-DIST(config-if)#tunnel destination 200.10.20.2
R-DIST(config-if)#mode gre ip
R-DIST(config-if)#exit
R-DIST(config)#ip route 172.16.1.0 255.255.255.0 10.0.99.3
R-DIST(config)#end
R-DIST#
Mar 01, 00:03:40.033: SYS-5-CONFIG_1: Configured from console by console
R-DIST#
R-DIST#

```

R6-VPN:

```

R6-VPN#
R6-VPN#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6-VPN(config)#hostname R6-VPN
R6-VPN(config)#interface Serial0/0/0
R6-VPN(config-if)#description LINK-R-DIST
R6-VPN(config-if)#ip address 200.10.20.2 255.255.255.252
R6-VPN(config-if)#no shutdown
R6-VPN(config-if)#exit
R6-VPN(config)#interface Loopback0
R6-VPN(config-if)#
R6-VPN(config-if)#ip address 172.16.1.1 255.255.255.0
R6-VPN(config-if)#no shutdown
R6-VPN(config-if)#exit
R6-VPN(config)#interface Tunnel0
R6-VPN(config-if)#
R6-VPN(config-if)#ip address 10.0.99.2 255.255.255.252
R6-VPN(config-if)#tunnel source Serial0/0/0
R6-VPN(config-if)#tunnel destination 200.10.20.1
R6-VPN(config-if)#mode gre ip
R6-VPN(config-if)#exit
R6-VPN(config)#ip route 192.168.10.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 192.168.20.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 192.168.30.0 255.255.255.0 10.0.99.1
R6-VPN(config)#ip route 0.0.0.0 0.0.0.0 200.10.20.1
R6-VPN(config)#end
R6-VPN#
Mar 01, 00:03:40.033: SYS-5-CONFIG_1: Configured from console by console
R6-VPN#
R6-VPN#

```

R-DIST e R6: Criação de um túnel GRE entre o R-DIST e o router remoto R6, com rotas estáticas para permitir comunicação entre as redes internas e o cliente VPN.

9) Teste e Validação

Realização de testes de ping entre filiais, verificação das tabelas de roteamento, análise do NAT, consulta das ACLs e do Port-Security, e validação do SSH com os utilizadores criados:

Testar ping entre filiais:

```
R3#ping 10.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 25/33/39 ms

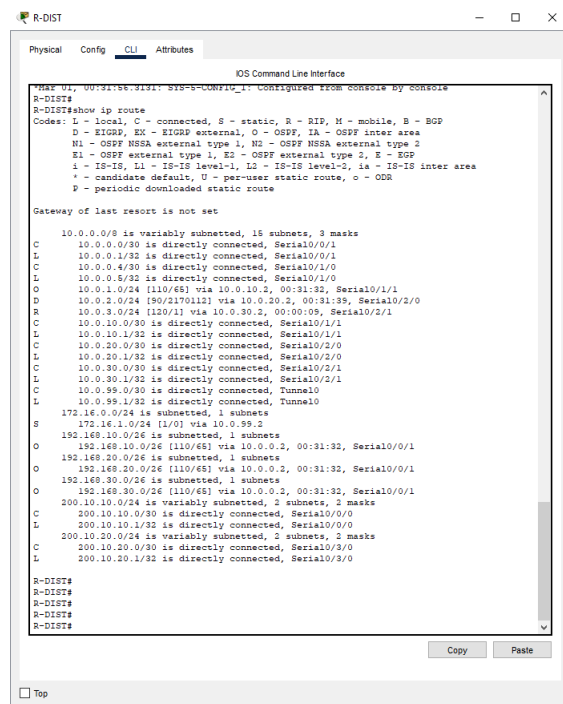
R4#ping 10.0.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 21/33/39 ms

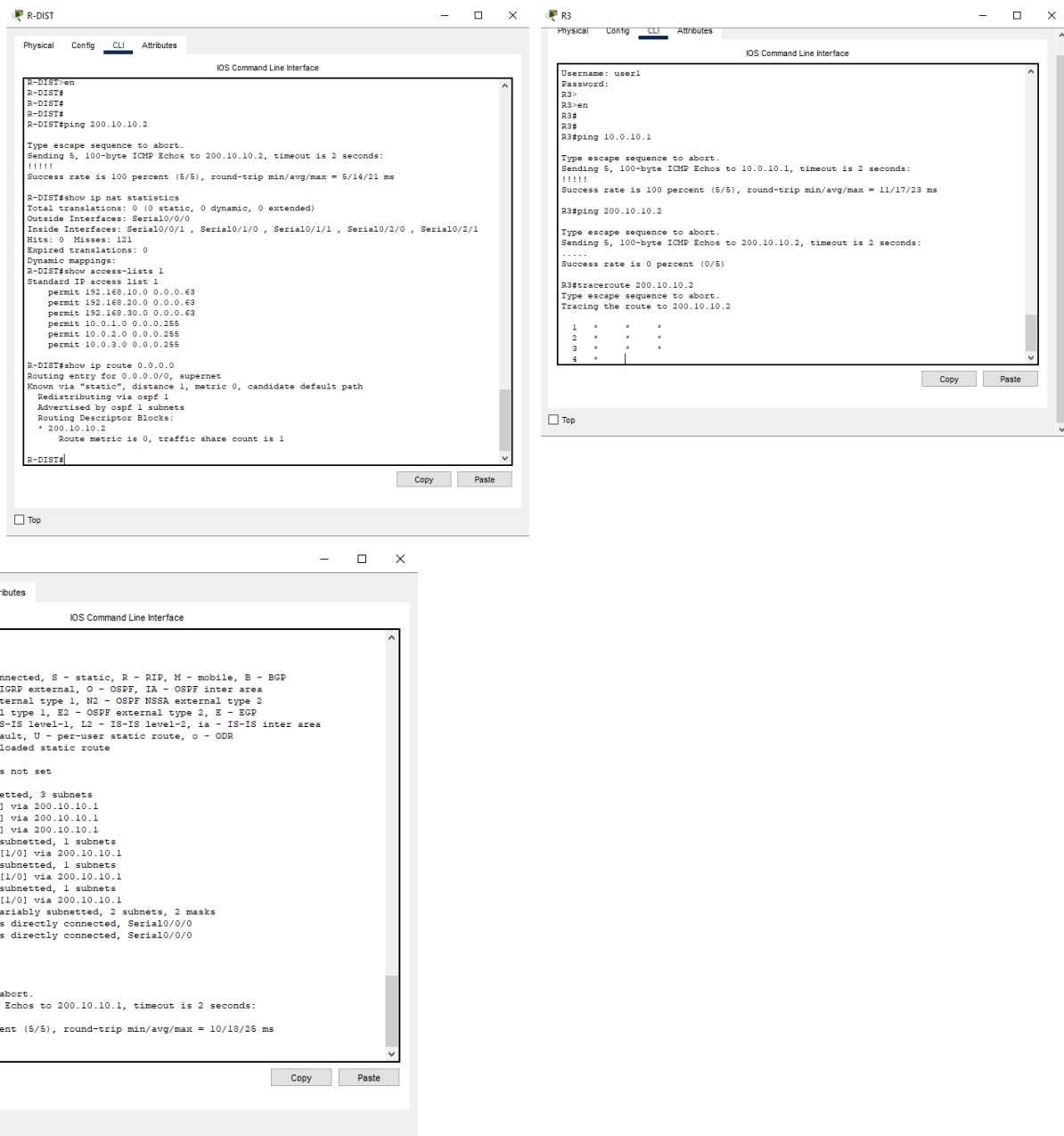
R5#ping 10.0.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 21/34/40 ms
```

Verificar redistribuição de rotas:



Validar NAT:



The image displays three screenshots of Cisco IOS Command Line Interface (CLI) windows, each showing the configuration and verification of NAT on a different router: R-DIST, R3, and ISP.

R-DIST Screenshot: The CLI shows the configuration of a static NAT pool named 'nat-internet' with IP addresses 192.168.10.0 to 192.168.10.255. The 'ip nat inside' command is configured on the 'Serial0/0/0' interface, and the 'ip nat outside' command is configured on the 'Serial0/0/1' interface. The 'ip nat enable' command is also shown. The output of the 'show ip nat translations' command is displayed, showing a successful translation of the internal IP 10.0.0.1 to the external IP 192.168.10.1.

R3 Screenshot: The CLI shows the configuration of a static NAT pool named 'nat-internet' with IP addresses 10.0.10.0 to 10.0.10.255. The 'ip nat inside' command is configured on the 'Serial0/0/0' interface, and the 'ip nat outside' command is configured on the 'Serial0/0/1' interface. The 'ip nat enable' command is also shown. The output of the 'show ip nat translations' command is displayed, showing a successful translation of the internal IP 10.0.10.1 to the external IP 10.0.10.1.

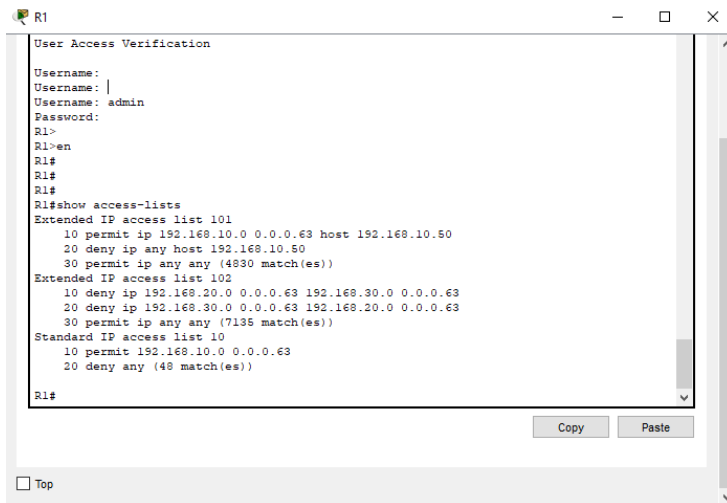
ISP Screenshot: The CLI shows the configuration of a static NAT pool named 'nat-internet' with IP addresses 10.0.0.0 to 10.0.0.255. The 'ip nat inside' command is configured on the 'Serial0/0/0' interface, and the 'ip nat outside' command is configured on the 'Serial0/0/1' interface. The 'ip nat enable' command is also shown. The output of the 'show ip nat translations' command is displayed, showing a successful translation of the internal IP 10.0.0.1 to the external IP 10.0.0.1.

A configuração do NAT foi efetuada: o R-DIST tem rota default, ACL 1 com todas as redes internas, e o tráfego chega à NAT (Misses: 101). O R3 alcança o R-DIST, e o ISP tem rotas de retorno e responde a pings do R-DIST.

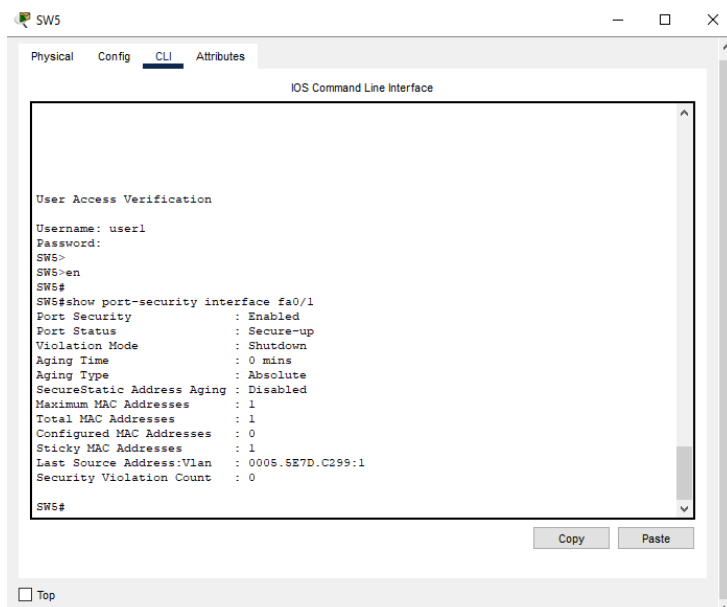
No entanto, o ping do R3 para o ISP falha e o traceroute mostra que o pacote não avança. O comando show ip nat translations mantém-se vazio, indicando que a tradução não ocorre.

Acredito que numa rede real, com equipamentos físicos, a configuração se comportaria de maneira diferente.

Confirmar ACLs e segurança de portas:



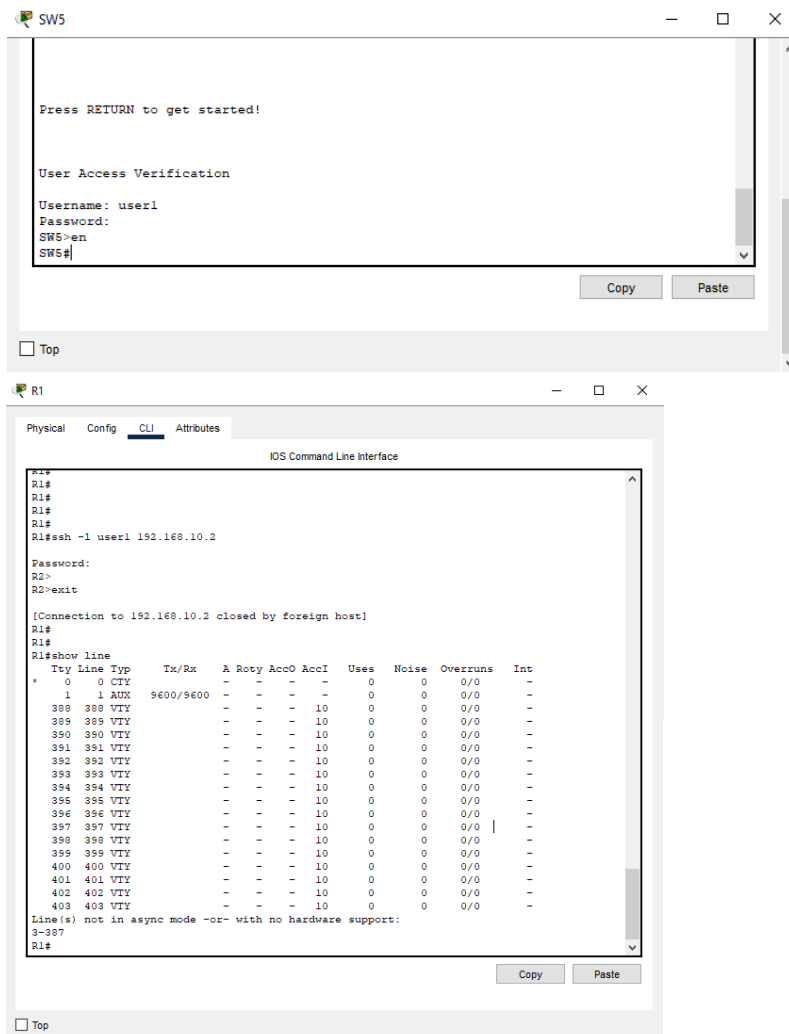
```
R1
User Access Verification
Username:
Username: admin
Password:
R1>
R1>en
R1#
R1#
R1#
R1#show access-lists
Extended IP access list 101
 10 permit ip 192.168.10.0 0.0.0.63 host 192.168.10.50
 20 deny ip any host 192.168.10.50
 30 permit ip any any (4830 match(es))
Extended IP access list 102
 10 deny ip 192.168.20.0 0.0.0.63 192.168.30.0 0.0.0.63
 20 deny ip 192.168.30.0 0.0.0.63 192.168.20.0 0.0.0.63
 30 permit ip any any (7135 match(es))
Standard IP access list 10
 10 permit 192.168.10.0 0.0.0.63
 20 deny any (48 match(es))
R1#
```



```
SW5
Physical Config CLI Attributes
IOS Command Line Interface

User Access Verification
Username: user1
Password:
SW5>
SW5>en
SW5#
SW5#show port-security interface fa0/1
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Shutdown
Aging Time : 0 mins
Aging Type : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 0
Sticky MAC Addresses : 1
Last Source Address:Vlan : 0005.5E7D.C299:1
Security Violation Count : 0
SW5#
```

Testar AAA e SSH:



```
SW5
Press RETURN to get started!

User Access Verification

Username: user1
Password:
SW5>en
SW5#

R1
Physical Config CLI Attributes
IOS Command Line Interface

R1#
R1#
R1#
R1#
R1#ssh -l user1 192.168.10.2

Password:
R2>
R2>exit

[Connection to 192.168.10.2 closed by foreign host]
R1#
R1#
R1#show line

  Tty Line Typ Tx/Rx  A Roty AccO AccI  Uses  Noise  Overruns  Int
*   0   0 CTY          -   -   -   -    0    0    0/0    -
  1   1 AUX 9600/9600  -   -   -   -    0    0    0/0    -
388 388 VTY          -   -   -  10    0    0    0/0    -
389 389 VTY          -   -   -  10    0    0    0/0    -
390 390 VTY          -   -   -  10    0    0    0/0    -
391 391 VTY          -   -   -  10    0    0    0/0    -
392 392 VTY          -   -   -  10    0    0    0/0    -
393 393 VTY          -   -   -  10    0    0    0/0    -
394 394 VTY          -   -   -  10    0    0    0/0    -
395 395 VTY          -   -   -  10    0    0    0/0    -
396 396 VTY          -   -   -  10    0    0    0/0    -
397 397 VTY          -   -   -  10    0    0    0/0    -
398 398 VTY          -   -   -  10    0    0    0/0    -
399 399 VTY          -   -   -  10    0    0    0/0    -
400 400 VTY          -   -   -  10    0    0    0/0    -
401 401 VTY          -   -   -  10    0    0    0/0    -
402 402 VTY          -   -   -  10    0    0    0/0    -
403 403 VTY          -   -   -  10    0    0    0/0    -
Line(s) not in async mode -or- with no hardware support:
3-387
R1#
```

Bibliografia

Fontes de informação usadas:

- Gravações das aulas,
- Trabalhos anteriormente feitos,
- Apontamentos.

Conclusão

A rede NETCORP ao ser implementada cumpriu a maioria dos requisitos propostos. A infraestrutura é funcional, segura e redundante, com comunicação total entre filiais e Data Center.

Durante o projeto, surgiram algumas limitações técnicas, nomeadamente ao 802.1X e o NAT, que não funcionaram conforme esperado.

Não consegui resolver tais erros pois não percebi se eram erros na minha configuração ou bugs da versão do Packet Tracer (Versão 8.2.2.0400).

Sempre que possível, foram implementadas alternativas para garantir o cumprimento dos objetivos do enunciado.

Este trabalho permitiu consolidar os conhecimentos adquiridos ao longo da UFCD, constituindo uma experiência de aprendizagem enriquecedora.